THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 1: **Chemical Reactions and Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. On strongly heating green ferrous sulphate crystals, the first change observed is: 1

(a) loss of water of crystallisation and green colour turning white
(b) formation of blue colour
(c) direct reddish-brown colour
(d) melting without change

2. In the balanced equation $Na_2CO_3 + HCl \longrightarrow NaCl + H_2O + CO_2$, the coefficient of HCl is: 1

(a) 3 **(b)** 2 **(c)** 1 **(d)** 4

3. $Pb(NO_3)_2 + 2\,KI \longrightarrow PbI_2 + 2\,KNO_3$ is a: 1

(a) double displacement (precipitation) reaction
(b) decomposition reaction
(c) displacement reaction
(d) combination reaction

4. Zinc reacting with dilute sulphuric acid makes the test tube warm. This shows the reaction is: 1

(a) endothermic **(b)** exothermic **(c)** a decomposition **(d)** a precipitation

5. Number of oxygen atoms on the product side of balanced $2\,C_2H_2 + 5\,O_2 \longrightarrow 4\,CO_2 + 2\,H_2O$ is: 1

(a) 8 **(b)** 5 **(c)** 12 **(d)** 10

6. Which of the following is NOT a redox reaction? 1

(a) $2\,Mg + O_2 \longrightarrow 2\,MgO$ **(b)** $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ **(c)** $CuO + H_2 \longrightarrow Cu + H_2O$ **(d)** $NaOH + HCl \longrightarrow NaCl + H_2O$

7. The rancid smell of old cooking oil is due to: 1

(a) reduction of oils **(b)** dissolution of air
(c) oxidation of fats and oils **(d)** evaporation

8. The sum of coefficients in balanced $NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O$ is: 1

(a) 4 **(b)** 7 **(c)** 5 **(d)** 6

9. The reaction $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ is an example of a: 1

(a) decomposition only **(b)** double displacement only
(c) displacement and redox reaction **(d)** combination and redox reaction

10. Which test distinguishes hydrogen gas? 1

(a) It turns lime water milky **(b)** It turns blue litmus red
(c) It rekindles a glowing splinter **(d)** It burns with a pop sound

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State two differences between corrosion and rancidity. 2

12. Balance the equation $Fe + H_2O \longrightarrow Fe_3O_4 + H_2$. 2

13. Dilute HCl is added to sodium carbonate. Write the balanced equation and name the gas. 2

14. Write the balanced equation for heating calcium carbonate and name the type of reaction. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Balance $\mathrm{Al + HCl \longrightarrow AlCl_3 + H_2}$ and name the gas given off. 3

16. Compare exothermic and endothermic reactions, giving one example for each. 3

17. Quick lime is added to water in a beaker. Give the observation, the equation and the type. 3

OR

Balance $\mathrm{H_3PO_4 + NaOH \longrightarrow Na_3PO_4 + H_2O}$ and identify the type. 3

18. Differentiate between painting and galvanisation as methods of preventing rusting. 3

OR

A student burns magnesium ribbon. Which precautions should she take and why? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Balance the following: 5

(i) $\mathrm{Fe + H_2O \longrightarrow Fe_3O_4 + H_2}$

(ii) $\mathrm{C_3H_8 + O_2 \longrightarrow CO_2 + H_2O}$

(iii) $\mathrm{KClO_3 \longrightarrow KCl + O_2}$.

OR

Explain the conservation of mass using the reaction of lead nitrate with potassium iodide in a closed flask. State the set-up, observation, and inference. 5

20. Write balanced chemical equations for: 5

(a) magnesium + hydrochloric acid

(b) potassium iodide + lead nitrate

(c) calcium hydroxide + carbon dioxide

(d) iron + steam

(e) heating of CaCO3.

OR

Describe the methods of preventing corrosion. Explain how galvanisation works and why alloying helps. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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